AN EXPLORATORY DESCRIPTIVE AUDIT OF MULTIPLE-CHOICE SCORING PROTOCOLS FOR LLADA-8B ON GSM8K-MC

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ABSTRACT

Multiple-choice accuracy for a diffusion language model depends on the scoring protocol as well as the model. We study this dependence in one bounded model–task setting: LLaDA-8B-Instruct with constructed four-choice GSM8K-MC pools. In the foregrounded matched-draw comparison, the source manuscript reports that a confidence-ordered ground-truth-injection scorer (`s_conf`) exceeds a uniform random-mask Monte Carlo reference (`mc_uniform`) by $13.0\,\mathrm{pp}$ on the $n{=}400$ pool. A separately identified, post-selected matched-forward-call-count sensitivity reports the same $13.0\,\mathrm{pp}$ gap, and a displayed 128-draw point reports $9.0\,\mathrm{pp}$. On a reported disjoint $n{=}819$ pool, fixed left-to-right injection is reported to exceed `s_conf` by $5.62\,\mathrm{pp}$ in aggregate, while the direction reverses at answer position 0. These are exploratory descriptive contrasts on named finite pools conditional on recorded scorer realizations. All empirical numbers are source-reported and cannot be independently verified from the current package; no experiment or analysis was rerun for this revision. The disclosed summaries describe setting-local differences among whole protocols, not an individual mechanism, stochastic-policy expectation, actual-cost comparison, or cross-bed law.

Evidence notation. Throughout, SR/CV means *source-reported; cannot be independently verified from the current package*. Every empirical numerical measurement in the abstract, prose, tables, and appendix has SR/CV status. Algebraic identities and declared protocol constants are not empirical measurements. The timestamped registration, complete hypothesis ledger, and complete defined-arm outcome matrix are absent. Accordingly, every reported analysis in this paper is exploratory and descriptive; no prospective, confirmatory, or complete-reporting status is claimed.

1 INTRODUCTION

A masked diffusion language model does not by itself determine a multiple-choice accuracy. A scorer must decide which candidate tokens remain masked, in what order tokens are revealed, whether known candidate tokens are fed back, how many model calls are used, and how token losses become one option score. Changing that tuple changes the measurement instrument.

We conduct an exploratory descriptive audit of the outcome subset currently disclosed for one LLaDA-8B-Instruct $\times$ GSM8K-MC model–task setting. The source defines five protocol tuples, but the available summaries report outcomes for only four; no outcome for `anti_inject` is disclosed. The contribution is not a new scorer, a complete defined-arm comparison, or a component-level causal result. It is a bounded organization of reported comparisons that (i) separates nominal draw matching from a forward-call-count sensitivity, (ii) conditions stochastic comparisons on the recorded realizations, and (iii) places a fixed-order aggregate next to its answer-position-0 reversal.

At first use below, `s_conf` means confidence-ordered ground-truth-token injection, `mc_uniform` means uniform random-mask Monte Carlo without injection, `rand_inject` means ground-truth injection in one recorded random reveal order, `fixed_inject` means ground-truth injection in fixed left-to-right order, and `anti_inject` means maximum-loss-first ground-truth injection; L_i denotes the candidate-token length. Table 1 is the canonical map of the disclosed results.

Table 1: Canonical map of disclosed results. All effects are SR/CV and exploratory descriptive quantities. Positive signs follow the contrast in column two. The source defines five protocols, but this table covers only the four with disclosed outcomes; it is not a prospective endpoint hierarchy.

Status	Explicit contrast	Pool / evidence ID	Reported effect	Interpretation boundary
Foregrounded descriptive	`s_conf` − `mc_uniform`, $n_{\mathrm{mc},i} = L_i$	$n = 400$; E11	+13.0 pp	Nominal scoring-draw match only
Post-selected sensitivity	`s_conf` − `mc_uniform`, $2L_i/2L_i$ calls	$n = 400$; E12	+13.0 pp	Forward-call-count sensitivity at one tested point
Displayed higher-budget point	`s_conf` − `mc_uniform`, $n_{\mathrm{mc}} = 128$	$n = 400$; E12	+9.0 pp	One reported point; no curve inference
One-schedule comparison	`s_conf` − `rand_inject`	fresh $n = 819$; E13	+12.58 pp	Conditional on one recorded random schedule
Fixed-order comparison	`fixed_inject` − `s_conf`	fresh $n = 819$; E14	+5.62 pp	Aggregate reverses at position 0

This map prevents two common overreads. First, the matched-L row is foregrounded because it matches nominal scoring draws, not because an accessible registration establishes prospective status. Second, neither the random-order nor fixed-order comparison identifies a general benefit of a reveal policy. Both are conditional finite-pool observations under the particular choice constructions described below.

2 Closest Work and Contribution Boundary

LLaDA introduced the masked-diffusion model family used here (Nie et al., 2025), and masked diffusion language modeling provides the broader objective and sampling context (Sahoo et al., 2024). The closest verified scorer source is iLLaDA (Nie et al., 2026). It defines a deterministic confidence-based multiple-choice surrogate that repeatedly reveals the ground-truth candidate token with greatest model confidence and sums its log probabilities. The paper explicitly treats that score as a task-specific surrogate rather than a likelihood estimate and compares it with a likelihood-style baseline on PIQA, ARC-Challenge, and HellaSwag. It does not report the present paired same-item audit, nominal-draw versus forward-call accounting, random-mask budget curve, reveal-order comparison, or GSM8K-MC construction analysis. Our contribution is therefore an audit of the measurement protocol, not the invention of confidence-guided scoring.

Prior work also shows that answer position can affect multiple-choice selection (Zheng et al., 2024). That result motivates caution around our position reversal but does not study masked-diffusion ground-truth-token injection or its call accounting. The current citation lock verifies these closest-work boundaries; a bibliography-wide metadata and sentence-support audit remains incomplete.

3 Study Definition

Bed and evidence boundary. The source manuscript describes one frozen LLaDA-8B-Instruct checkpoint and four-choice pools derived from GSM8K (Cobbe et al., 2021). It reports two historical balanced pilots, an exactly answer-position-balanced $n = 400$ pool, and a disjoint fresh $n = 819$ pool with approximate position balance. The fresh pool mixes 336 near-miss items and 483 items whose distractors were drawn from a fixed answer pool. Arms are paired within each named pool and use the same recorded choice set. The current package contains neither item identifiers nor construction code, so pool membership, disjointness, prompts, and choice construction cannot be independently checked.

The source describes BF16 inference without generation or temperature. Confidence and fixed schedules are deterministic. The random-order result uses one stored schedule realization; the random-mask reference uses one stored collection of mask draws at each reported budget. The underlying orders and draws are not in the current package. Method statements in this section are therefore manuscript-described rather than code-verified.

Defined protocols and disclosed-outcome boundary. For candidate tokens $y_{1:L_i}$, an injected arm reads a token loss, reveals the known candidate token at a selected position, feeds that token into later steps, and sums the per-step losses. `s_conf` selects the minimum-loss position, `anti_inject` the maximum-loss position, `rand_inject` a seeded-random order, and `fixed_inject` a fixed left-to-right order. The non-injected `mc_uniform` arm samples a masked subset of size $k \sim \mathrm{U}\{1, \ldots, L_i\}$ on each draw, rescales the partial-mask loss by L_i/k, and averages across draws. All arms select the option with minimum reported score, with first-index tie breaking. These are five source-defined protocols. The disclosed outcome summaries used below cover `s_conf`, `mc_uniform`, `rand_inject`, and `fixed_inject`; the current package contains no reported `anti_inject` outcome, so this paper does not present a complete five-protocol comparison.

Table 2: Manuscript-described decomposition of the five defined protocols. Only four have disclosed outcome summaries in this paper. The implementation and call traces are absent, so the accounting cannot be independently verified.

Arm	Reveal schedule	GT feedback	Reported calls per candidate
`s_conf`	minimum loss first	yes	L_i score $+L_i$ select
`rand_inject`	recorded random order	yes	L_i
`anti_inject`	maximum loss first	yes	L_i score $+L_i$ select
`fixed_inject`	fixed left-to-right	yes	L_i
`mc_uniform`	random masked subsets	no	n_{mc}

The paper uses a *forward call* only as the source manuscript's model- invocation accounting unit. Equality in that count does not establish equality of FLOPs, latency, memory, throughput, or energy. This is the only resource-accounting claim made in the paper; no broader equivalence is implied.

Finite-pool, realized-schedule estimand. Let $\mathcal{P}$ be one named finite pool and let ω denote the complete recorded realization of every stochastic scoring choice: random-mask subsets for `mc_uniform` and reveal orders for `rand_inject`. For deterministic arms, ω is empty. The target is

$$\Delta_{\mathcal{P},\omega}(A, B) = \frac{1}{|\mathcal{P}|} \sum_{q \in \mathcal{P}} \left[\nVdash\{A_\omega(q) = y_q\} - \nVdash\{B_\omega(q) = y_q\} \right]. \tag{1}$$

It is a descriptive contrast for the realized item set and scorer traces, not an expectation over a source population, mask streams, or random schedules. For a printed contrast $A - B$, b counts items on which A is correct and B is incorrect, while c counts the reverse direction; hence $\widehat{\Delta} = (b - c)/n$ and $m = b + c$ is the total discordant count. These are arithmetic summaries of the fixed realized record. Because the declared target supplies no item-sampling, assignment, or scorer-randomness model, this paper omits the source manuscript's confidence intervals, p-values, and cross-pool test and gives them no inferential role. The absent registration and complete outcome ledger likewise preclude prospective or prespecified- family labels.

4 Descriptive Reference-Budget Comparisons

Table 3 reports the foregrounded matched-L comparison and three other displayed budget points on the same named $n = 400$ pool. E11 designates the missing matched-L record; E12 designates the missing $2L$ and 128-draw records; E10 designates the missing single-draw record. These are opaque paper-local IDs, not resolvable artifacts in the current package, and no crosswalk is available.

Table 3: SR/CV exploratory descriptive reference-budget contrasts on the reported $n = 400$ pool. For each printed s_conf-minus-reference contrast, b counts s_conf-only correct items and c reference-only correct items. No interval or test is used.

Reference budget (ref./s_conf reported calls)	n	Acc. pair (s_conf/ref.)	b/c	$\widehat{\Delta}$	Status
Single draw ($1/2L$)	400	42.75/28.75%	113/57	+14.0 pp	displayed point (E10)
$n_{\mathrm{mc},i} = L_i$ ($L/2L$)	400	42.75/29.75%	106/54	+13.0 pp	foregrounded descriptive (E11)
$n_{\mathrm{mc},i} = 2L_i$ ($2L/2L$)	400	42.75/29.75%	109/57	+13.0 pp	post-selected sensitivity (E12)
$n_{\mathrm{mc}} = 128$ ($128/2L$)	400	42.75/33.75%	82/46	+9.0 pp	displayed higher-budget point (E12)

At the foregrounded matched-L point, both arms receive L_i nominal scoring draws, but the manuscript ledger assigns s_conf another L_i schedule-selection calls. The source-reported finite-pool gap is $+13.0$ pp. The post-selected $2L/2L$ row has the same reported gap but a different discordance record. Conditional on the manuscript's unverified call ledger, this sensitivity addresses only the explanation that the reference is lower because it receives fewer counted forward calls at that tested point. It neither becomes a second headline result nor isolates feedback, reveal order, or estimator choice.

The 128-draw row has the smallest reported gap on the displayed curve. The reference changes with budget, so the four rows are operating points of one protocol replacement rather than independent replications. Nothing here supports a dose-response form or behavior between or beyond the tested budgets.

5 Descriptive Reveal-Order Results

Reveal-order evidence addresses two different questions and is reported in two directions. Table 4 uses s_conf minus rand_inject and is conditional on one recorded random schedule. Table 5 uses fixed_inject minus s_conf throughout, so a positive value always favors fixed left-to-right reveal.

5.1 One recorded random schedule

Table 4: SR/CV contrasts for s_conf minus rand_inject. These rows concern one recorded schedule realization, not the expectation over random schedules. Here $m = b + c$ is the total number of discordant items for the printed contrast. The pooled row combines different named pools and is only a disclosure.

Pool	n	$\widehat{\Delta}$	m	Status
Fresh	819	+12.58 pp	305	post-selected exploratory (E13)
$n = 400$ pilot	400	+3.75 pp	141	undetermined
Pooled disclosure	1219	+9.68 pp	446	exploratory; not a common effect

The fresh and pilot rows report different descriptive magnitudes under the stored schedules. Because the pools differ and the declared target provides no cross-pool probability model, the pooled row is not interpreted as a common effect and no cross-pool test is used. We also do not average over schedules or infer that confidence ordering is generally better than random ordering. Multi-schedule evaluation remains an evidence blocker.

5.2 Fixed left-to-right reveal and the position reversal

Table 5: SR/CV contrasts for `fixed_inject` minus `s_conf`; positive values favor fixed order. The position-0 row is adjacent to the aggregate so its reversal is explicit. Here b counts fixed-only correct items and c confidence-only correct items; no interval or test is used.

Scope	n	b/c	$\widehat{\Delta}$	Status
Fresh aggregate	819	145/99	$+5.62$ pp	exploratory aggregate (E14)
Fresh, answer position 0	199	21/46	-12.56 pp	post-hoc subgroup (E14)

The fresh aggregate favors fixed order descriptively, whereas the answer-position-0 subgroup favors `s_conf`. Construction-stratified summaries also differ: the source reports fixed-order margins of 2.98 pp in the near-miss stratum and 7.45 pp in the answer-pool stratum (Appendix Table 6). The position analysis is post-hoc, and no tie-robust interaction analysis is available. The opposite signs describe the realized fresh-pool record; they do not establish a general answer-position mechanism or show that confidence ordering is generally unnecessary.

6 Limitations, Availability, and Disclosure

Scientific scope. The study covers one checkpoint and constructed four-choice pools from one task. The foregrounded protocol swap changes feedback, masking path, estimator, and reported call count together. The random-order result conditions on one unavailable schedule trace, while the fixed-order result depends on construction and answer position. No displayed descriptive quantity identifies a component cause or an expectation over scorer randomness. Cross-model, cross-task, multi-seed, and independently constructed-pool behavior remain unmeasured here.

Unresolved evidence blockers. The current package cannot resolve three evidence-dependent actions: recovery or replay of the central paired records and exact protocol implementation; disclosure of the complete evaluated-arm outcome matrix and decision chronology; and multi-realization evaluation plus a disjoint-pool replication of the matched-L comparison. No missing artifact was reconstructed, and no experiment or new analysis was run to address these blockers.

What the current package contains. The authoring directory inspected for this revision contains `main.tex`, `appendix_refine5.tex`, `refs.bib`, the local ICLR/BibTeX and vendored style files, and current build products. It also contains seven legacy pre-rendered PDF/PNG figure assets but no figure-generation script; none of those assets is included by the current LaTeX source. The directory does not contain the item-level experimental records, recorded schedules, machine-readable registrations, protocol or analysis code, environment lock, or an ID-to-file crosswalk required to verify E10–E15. Thus the empirical values were neither rerun nor independently recomputed for this revision.

AI assistance disclosure. AI tools assisted with language editing, LaTeX refactoring, and isolated internal pre-submission review for this revision. The authors remain responsible for the manuscript's claims and evidence boundaries. AI assistance did not provide experimental records, generate new measurements, or verify the source-reported results.

7 Conclusion

In this exploratory descriptive audit, the foregrounded matched-L `s_conf–mc_uniform` row reports $+13.0$ pp on the named $n = 400$ pool. The $2L/2L$ row is a post-selected forward-call-count sensitivity, and the 128-draw row is one displayed higher-budget point. Separately, fixed order is reported to have higher aggregate accuracy on the fresh pool while having lower accuracy in the answer-position-0 subgroup. Conditional on the named pools and recorded scorer realizations, these disclosed summaries describe nonidentical protocol outputs; they do not establish a general ordering. Missing records, complete-arm outcomes, chronology, implementations, and multi-realization evidence prevent stronger empirical or causal conclusions.

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A EVIDENCE BOUNDARY AND POOL DESCRIPTION

Measurement status. The global SR/CV definition in the main text applies to every empirical number below. The appendix adds no reconstructed record, rerun, or new measurement.

Finite pools and pairing. The source manuscript describes an exactly answer-position-balanced $n = 400$ four-choice pool and a disjoint fresh $n = 819$ four-choice pool with approximate position balance. All arms reportedly score the same recorded choice set within a pool, so the accuracy contrasts are paired by question. The first 336 fresh items use near-miss distractors; the remaining 483 use three distinct values drawn from a fixed answer pool. Neither the item manifest nor the distractor-construction implementation is present. The finite-pool estimand in Equation 1 therefore conditions on the named pools exactly as reported and does not assert a sampling mechanism or population target.

Stochastic scorer realizations. The source says that `rand_inject` uses one persisted schedule and that `mc_uniform` uses recorded mask draws at each budget. It also reports that the original random schedule depended on a process-sensitive string hash, making the nominal seed insufficient for replay. The per-candidate schedules and mask draws are absent from the package. Consequently, the random-order and random-mask results are conditional on unavailable realized traces, and a schedule-family or mask-stream expectation is unidentified.

B FIXED-ORDER CONSTRUCTION AND POSITION DETAIL

Table 6 uses one direction throughout: `fixed_inject` minus `s_conf`. Positive values favor fixed left-to-right reveal; negative values favor confidence ordering. This orientation is the same as Table 5.

Table 6: SR/CV fresh-pool fixed-order contrasts, consistently oriented as `fixed_inject` minus `s_conf`. The construction split and answer-position split are post-hoc re-presentations of the source manuscript's aggregate record. The position rows do not constitute a formal interaction analysis. For every row, b counts fixed-only correct items and c confidence-only correct items.

Scope	n	b/c	$\widehat{\Delta}$	Reading
Near-miss construction	336	61/51	$+2.98$ pp	fixed direction
Answer-pool construction	483	84/48	$+7.45$ pp	fixed direction
Answer position 0	199	21/46	-12.56 pp	reversal: `s_conf` direction
Answer position 1	207	37/15	$+10.63$ pp	fixed direction
Answer position 2	226	44/18	$+11.50$ pp	fixed direction
Answer position 3	187	43/20	$+12.30$ pp	fixed direction

The score vectors needed for tie-robust or construction-adjusted interaction analyses are not present. Accordingly, the main text reports the sign reversal descriptively and does not claim a general answer-position mechanism or give the source manuscript's inferential quantities any role.

C Reported Budget and Length Accounting

Protocol tuple. The whole-protocol interventions differ along several axes. The foregrounded matched-L comparison changes feedback, masking path, estimator, and the reported number of selection calls while matching nominal scoring draws. The $2L/2L$ post-selected sensitivity retains those protocol changes and matches only the source manuscript's count of model invocations. The fixed-order and random-order comparisons retain ground-truth feedback but change reveal order; confidence ordering also carries reported schedule-selection calls. None of these comparisons isolates a single component.

Table 7: Manuscript-described per-candidate accounting. These are protocol definitions and reported call counts, not measured resource costs. The implementation and instrumented call traces are absent.

Protocol	Scoring draws	Selection calls	Total calls	Role
`s_conf` / `anti_inject`	L	L	$2L$	injected, model-derived order
`rand_inject` / `fixed_inject`	L	0	L	injected, recorded random or fixed order
`mc_uniform` single draw	1	0	1	displayed budget point
`mc_uniform` matched-L	L	0	L	foregrounded descriptive budget
`mc_uniform` matched-$2L$	$2L$	0	$2L$	post-selected forward-call sensitivity
`mc_uniform` higher budget	128	0	128	displayed budget point

Table 8: SR/CV candidate-length counts transcribed from the source manuscript. Each L-bin entry counts candidate strings, not questions, so the row denominator is $4n$; mean L is likewise over candidate strings. The records needed to verify tokenization and these distributions are absent.

Pool	Items	$L=3$	$L=4$	$L=5$	$L=6$	$L=7$	$L=8$	$L=9$	Mean L
Historical	64	39	120	72	17	8	0	0	4.355
$n=400$ reference pool	400	262	751	424	118	30	15	0	4.343
Fresh pool	819	364	1237	1010	390	167	101	7	4.722

Inferential quantities omitted. The source manuscript reports intervals, paired tests, a cross-pool test, and a multiple-testing family. The declared fixed-pool, realized-trace target supplies no probability model for interpreting those quantities, and the underlying records, code, complete hypothesis list, and chronology are absent. They are therefore omitted from the result tables and support no conclusion in this revision. Every retained numerical result is a descriptive effect, accuracy, count, or protocol-accounting value with SR/CV status.

D OPAQUE EVIDENCE-ID MANIFEST

Table 9 uses only opaque paper-local IDs. No ID resolves to a file in the current package, and the confidential crosswalk was not available for this revision.

Table 9: Required evidence objects. Quantitative scope values are SR/CV; every object and the ID crosswalk are absent from the current package.

ID	Intended object	Current status
E10	Single-draw paired record, reported $n = 400$	missing
E11	Matched-L foregrounded paired record, reported $n = 400$	missing
E12	$2L/2L$ sensitivity and 128-draw displayed-point records, reported $n = 400$	missing
E13	Fresh `s_conf/rand_inject` record and stored schedule, reported $n = 819$	missing
E14	Fresh `s_conf/fixed_inject` record with construction, position, and tie fields, reported $n = 819$	missing
E15	Reference chance-calibration ledger	missing

Table 10: Minimum contents of a complete anonymous experimental bundle. This is a requirements checklist, not evidence that the components are present.

Component	Required contents	Independent check
Sampling frames	Item IDs, split membership, exclusions, pool assignment, and overlap report	Verify pool sizes, balance, and disjointness
Choice construction	Distractor rules, deduplication, shuffling, answer positions, and candidate strings	Rebuild and byte-compare every choice set
Prompts and model	Prompt bytes; model and tokenizer revisions; precision and environment	Reconstruct tokenization and scoring inputs
Protocol traces	Implementations, configurations, tie policy, mask draws, and reveal orders	Replay each arm and verify score vectors and predictions
Statistics	Paired records, descriptive aggregation code, complete evaluated-arm ledger, and decision chronology	Recompute every displayed effect and count; verify exploratory labels
Manifest	Content hashes, environment lock, and canonical command sequence	Validate the bundle before recomputation

Package cross-reference. Section 6 states the exact authoring-package boundary. Seven legacy pre-rendered figure assets remain in the directory, but no figure is included and no generation script or E10–E15 object is present.

E CITATION-AUDIT BOUNDARY

The citation lock verifies the primary-source metadata and local support used for the closest-work boundaries of iLLaDA, LLaDA, the multiple-choice position-sensitivity study, and masked diffusion language modeling (Nie et al., 2026; 2025; Zheng et al., 2024; Sahoo et al., 2024). In particular, iLLaDA supplies the closest verified scoring mechanism but not this paper's paired finite-pool protocol and budget audit. The lock does not claim a complete bibliography-wide audit. Remaining entries and broad background statements must be checked against primary sources before a full citation-verification claim can be made.